

Priyanka Vijaybaskar

412-224-8538 | pvijayba@andrew.cmu.edu | www.linkedin.com/in/priyankavijaybaskar | <https://github.com/prinks-wiz> | <https://prinks-wiz.github.io/>

Summary

Physical AI engineer with experience shipping end-to-end ML systems, from data acquisition pipelines, model training, and inference to optimized real-time deployment on edge GPUs with CUDA, ONNX, and TensorRT.

EDUCATION

Carnegie Mellon University

Aug 2025 - Dec 2026

Master's, Computational Data Science (GPA: 3.85/4.00)

Pittsburgh, PA

• **Coursework:** Cloud Computing, Neuro Symbolic AI, Multimodal AI, Visual-Learning Recognition

Sri Sivasubramaniya Nadar College of Engineering, Anna University

Sep 2021 - Jun 2025

Bachelors of Technology, Information Technology (GPA: 8.12/10)

Chennai, India

• **Achievements:** President, ProCoDe.

EXPERIENCE

Argus Spacecraft (CMU) | Graduate Researcher (Flight Software-Payload)

Oct 2025 - Present

- Replicated advanced **visual-localization** research end-to-end and **open-sourced the full pipeline**, creating a **2.7 M+tile** georeferenced **Sentinel-2 multispectral reference** database at flight-matched resolution with **800 K astronaut photographs** as queries, enabling rapid on-board localization for mission planning
- **Developed training strategies** for vision transformers (DINOv2 with LoRA fine-tuning) across query and reference databases for cross-view image-to-map localization, improving **top-match retrieval accuracy by 67%**.
- Built geometric matching algorithms with **SIFT** and **LightGlue** to identify **unique patterns** among visually similar terrain regions, strengthening inlier verification for **pose estimation**. Applied **state estimation to fuse visual localization** with an orbit-determination solver, using image-segmented landmarks and inliers to jointly estimate 6-DOF position, velocity, and orbital dynamics from monocular onboard imagery, **reducing velocity error** to about **13 m/s in multi-frame evaluation**.
- Optimized the **perception stack** for edge deployment on **NVIDIA Jetson Orin** under onboard sensor and compute constraints, achieving a **700x speedup** with custom **CUDA kernels** and **65% model compression** with **40% lower latency** via **ONNX**, **TensorRT**, **INT8 quantization**, and structured pruning within a **3GB RAM budget**.

Finsire Technologies | Data Science Intern

Jan 2024 - Apr 2024

- Engineered **time-series forecasting** model using Continuous Wavelet Transform for feature engineering on **500K+ mutual fund records** from MFI API; implemented hyperparameter optimization improving prediction **accuracy 15%** over baseline LSTM, directly securing 3 institutional clients.
- Built automated ETL pipelines for data extraction of **2K real estate listings** and SEBI financial statements; developed TF-IDF semantic matching algorithm with processing 50K+ vehicle-insurance pairs achieving **92% precision** for deployment.

PROJECTS

Cloud-Native Twitter Recommendation Service on AWS

Jan 2026 - May 2026

Carnegie Mellon University

Pittsburgh, PA

- Designed and deployed a **Kubernetes microservice architecture** on **AWS** using **kOps**, serving **18,000 RPS** at **93.5% correctness** across a heterogeneous cluster of ARM-based Graviton instances.
- Delivered a **41.7x throughput improvement** through **schema denormalization (3.5x)** with **Go** for faster inference, **InnoDB buffer pool** tuning to a 98.4% cache hit ratio, prepared statement caching, and a **45-second auth token TTL** that cut authentication overhead by 95%.
- Migrated from a self-hosted MySQL server to **Amazon Aurora with Redis and Memcached caching layers**, **gRPC-based authentication**, NLB routing, AWS auto-scaling and load balancing, and persistent **S3 storage**, then automated delivery with a **GitHub Actions** and **Helm CI/CD pipeline** using SHA-tagged deployments and database snapshots to **recreate the full cluster in under 8 minutes**.

Data Influence for Vision-Language Models (VLM)

Feb 2026 - Present

Carnegie Mellon University

Pittsburgh, PA

- Fine-tuned VLMs via **LoRA** (training <2% of 7B parameters) with **7 A100-targeted optimizations** including gradient checkpointing, TF32 precision, and 8-bit AdamW, reducing training time from days to hours on a single 40GB GPU.
- Evaluated **vision-language** and **vision-language-action** models for **autonomous driving action prediction**, curating driving datasets to surface anomalies and failure modes in scene understanding.
- Improved action prediction in **VLA by 36%** through **supervised fine-tuning (SFT)** and by **25% through optimized LoRA** fine-tuning of 7B models, using gradient checkpointing, **TF32 precision**, and **8-bit AdamW to train on a single 40GB A100**.

Physics-Based Motion Prediction using Neural ODE-VAE Architecture

Sep 2025 - Dec 2025

Carnegie Mellon University

Pittsburgh, PA

- Benchmarked multiple generative models (GANs, Pix2pix, diffusion models, VLM fine-tuning) for **trajectory prediction** on PhysicsGen dataset; designed novel Neural ODE-VAE hybrid architecture improving upon **baseline by 75%**.
- Implemented dynamics modelling with Neural ODEs integrated into VAE latent space, enabling physically-consistent trajectory generation of images.

SKILLS

- **Frameworks and Libraries:** Tensorflow, Numpy, Keras, Pandas, Seaborn, Matplotlib, Langchain, LangGraph, Flask, PyTorch, FastAPI, Docker, Vertex AI, Azure, GCP, AWS, CI/CD, RAG, GraphRAG, OpenCV, AWS, Kubernetes, Maven, Software Engineering, CUDA, Multimodal AI, Computer Vision, Helm, Agentic AI, State estimation
- **Programming Languages:** Python, SQL, Java, JavaScript, C, C++, Rust, Go, Typescript
- **Tools:** Git, NVIDIA Jetson, Weaviate, FAISS, W&B, TensorBoard, Azure, GCP, CPU Architecture, Agile